

```
mysql> CREATE DATABASE ecommerce;
Query OK, 1 row affected (0.02 sec)

mysql> USE ecommerce;
Database changed
mysql> CREATE TABLE customers (
  ->     customer_id INT PRIMARY KEY,
  ->     customer_name VARCHAR(100),
  ->     city VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE products (
  ->     product_id INT PRIMARY KEY,
  ->     product_name VARCHAR(100),
  ->     price DECIMAL(10,2)
  -> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE orders (
  ->     order_id INT PRIMARY KEY,
  ->     customer_id INT,
  ->     product_id INT,
  ->     quantity INT,
  ->     order_date DATE
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO customers VALUES
  -> (1, 'Rahul', 'Delhi'),
  -> (2, 'Priya', 'Mumbai'),
  -> (3, 'Amit', 'Jaipur'),
  -> (4, 'Neha', 'Pune');
Query OK, 4 rows affected (0.02 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> INSERT INTO products VALUES
```

```
mysql> INSERT INTO products VALUES
-> (101, 'Laptop', 50000),
-> (102, 'Mobile', 20000),
-> (103, 'Headphones', 3000),
-> (104, 'Keyboard', 1500);
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO orders VALUES
-> (1, 1, 101, 1, '2025-07-01'),
-> (2, 2, 102, 2, '2025-07-02'),
-> (3, 3, 103, 3, '2025-07-03'),
-> (4, 1, 104, 2, '2025-07-04'),
-> (5, 4, 102, 1, '2025-07-05');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM customers;
+-----+-----+-----+
| customer_id | customer_name | city |
+-----+-----+-----+
| 1 | Rahul | Delhi |
| 2 | Priya | Mumbai |
| 3 | Amit | Jaipur |
| 4 | Neha | Pune |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM products;
+-----+-----+-----+
| product_id | product_name | price |
+-----+-----+-----+
| 101 | Laptop | 50000.00 |
| 102 | Mobile | 20000.00 |
| 103 | Headphones | 3000.00 |
| 104 | Keyboard | 1500.00 |
+-----+-----+-----+
```

```
mysql> SELECT * FROM orders;
```

order_id	customer_id	product_id	quantity	order_date
1	1	101	1	2025-07-01
2	2	102	2	2025-07-02
3	3	103	3	2025-07-03
4	1	104	2	2025-07-04
5	4	102	1	2025-07-05

```
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM products  
-> WHERE price > 5000;
```

product_id	product_name	price
101	Laptop	50000.00
102	Mobile	20000.00

```
2 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM products  
-> ORDER BY price DESC;
```

product_id	product_name	price
101	Laptop	50000.00
102	Mobile	20000.00
103	Headphones	3000.00
104	Keyboard	1500.00

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT customer_id,  
-> SUM(quantity) AS total_quantity  
-> FROM orders
```

```
mysql> SELECT customer_id,  
-> SUM(quantity) AS total_quantity  
-> FROM orders  
-> GROUP BY customer_id;
```

customer_id	total_quantity
1	3
2	2
3	3
4	1

4 rows in set (0.00 sec)

```
mysql> SELECT c.customer_name,  
-> p.product_name,  
-> o.quantity  
-> FROM orders o  
-> INNER JOIN customers c  
-> ON o.customer_id = c.customer_id  
-> INNER JOIN products p  
-> ON o.product_id = p.product_id;
```

customer_name	product_name	quantity
Rahul	Laptop	1
Priya	Mobile	2
Amit	Headphones	3
Rahul	Keyboard	2
Neha	Mobile	1

5 rows in set (0.00 sec)

```
mysql> SELECT c.customer_name,  
-> o.order_id  
-> FROM customers c  
-> LEFT JOIN orders o
```



```
mysql> SELECT c.customer_name,  
-> o.order_id  
-> FROM customers c  
-> LEFT JOIN orders o  
-> ON c.customer_id = o.customer_id;
```

customer_name	order_id
Rahul	4
Rahul	1
Priya	2
Amit	3
Neha	5

5 rows in set (0.00 sec)

```
mysql> SELECT c.customer_name,  
-> o.order_id  
-> FROM customers c  
-> RIGHT JOIN orders o  
-> ON c.customer_id = o.customer_id;
```

customer_name	order_id
Rahul	1
Priya	2
Amit	3
Rahul	4
Neha	5

5 rows in set (0.00 sec)

```
mysql> SELECT *  
-> FROM products  
-> WHERE price >  
-> (  
-> SELECT AVG(price)  
-> FROM products
```

```
mysql> SELECT *  
-> FROM products  
-> WHERE price >  
-> (  
-> SELECT AVG(price)  
-> FROM products  
-> );
```

product_id	product_name	price
101	Laptop	50000.00
102	Mobile	20000.00

2 rows in set (0.00 sec)

```
mysql> SELECT SUM(quantity) AS total_quantity  
-> FROM orders;
```

total_quantity
9

1 row in set (0.00 sec)

```
mysql> SELECT AVG(price) AS average_price  
-> FROM products;
```

average_price
18625.000000

1 row in set (0.00 sec)

```
mysql> CREATE VIEW sales_summary AS  
-> SELECT customer_id,  
-> SUM(quantity) AS total_sales  
-> FROM orders
```

```
+-----+
|          9 |
+-----+
```

1 row in set (0.00 sec)

```
mysql> SELECT AVG(price) AS average_price
-> FROM products;
```

```
+-----+
| average_price |
+-----+
| 18625.000000 |
+-----+
```

1 row in set (0.00 sec)

```
mysql> CREATE VIEW sales_summary AS
-> SELECT customer_id,
-> SUM(quantity) AS total_sales
-> FROM orders
-> GROUP BY customer_id;
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> SELECT * FROM sales_summary;
```

```
+-----+-----+
| customer_id | total_sales |
+-----+-----+
|          1 |          3 |
|          2 |          2 |
|          3 |          3 |
|          4 |          1 |
+-----+-----+
```

4 rows in set (0.00 sec)

```
mysql> CREATE INDEX idx_customer
-> ON orders(customer_id);
```

Query OK, 0 rows affected (0.05 sec)

Records: 0 Duplicates: 0 Warnings: 0

```
mysql> |
```